AKNLTP014_REMED1_RCB Benchmark Parity Report

Input: /Users/arunsasidharan/EEGdata/EEGAnalysisCode/sampleData/AKNLTP014_REMED1_RCB.set

Scope: Rust app output compared against benchmark reference outputs for feature extraction and preprocessing.

Benchmark Sources

GEDAI benchmark source:
/Users/arunsasidharan/Code/ActiveProjects/ABRL_OctaveMatlab_tools/external_functions/GEDAI

FOOOF/specparam benchmark source: /Users/arunsasidharan/Code/Python_misc/fooof-main

MATLAB and Octave executables were not available on PATH in this shell, so the GEDAI benchmark source is documented and the Rust GEDAI implementation is aligned to the MATLAB algorithmic reconstruction path. The numerical preprocessing comparison plot is against the local reproducible reference pipeline artifact generated for AKNL.

FOOOF parity is evaluated against the local specparam/FOOOF algorithm source path requested above, through the same Welch PSD inputs used by the app report harness.

Feature Parity Summary

Python/reference feature shape: [10, 111]

Rust feature shape: [10, 111]

Common numeric features compared: 104

Family	Features	Max abs	Median mean abs	Median corr
acw	1	0	0	1
connectivity	54	0.67517	2.68137e-09	1
fooof	19	5.57945	0.000710271	0.99998
irasa	12	1.33674e-06	2.24617e-08	1
metadata	3	0	0	NA
nonlinear	8	0.000319949	6.53261e-10	1
psd	7	2.05291e-08	3.89176e-09	1

Targeted Remaining Metrics

Metric	Max abs	Mean abs	Corr
auc_FOOOF	5.57945	1.40531	0.994337
intercept_Irasa	1.33674e-06	5.89644e-07	1
slope_Irasa	6.35226e-07	2.98971e-07	1
rsquared_Irasa	6.64319e-08	4.48781e-08	1
auc_Irasa	5.41146e-07	2.42293e-07	1
oscspectraledge_Irasa	0	0	1
conn_mic_Theta	0.220766	0.220766	NA
conn_mic_ThetaAlpha	0.216948	0.216948	1

Metric	Max abs	Mean abs	Corr
conn_mic_Alpha	0.67517	0.67517	NA
conn_mic_Beta1	0.218856	0.218856	NA
conn_mic_Beta2	0.125749	0.125749	1
conn_mic_Gamma1	0.364188	0.364188	NA

Preprocessing Snapshot Summary

Python/reference sample rate: 250.0

Rust sample rate: 250.0

Channels compared: Fz, F3, F4, C3, C4, P3, P4, Pz, O1, O2

Samples compared: 1000

RMS difference (uV): 4.33829

Median absolute difference (uV): 2.8171

Max absolute difference (uV): 22.1237

Rust GEDAI SENSAI score: 14.6238

Rust GEDAI thresholds: [10.0, 0.0]

Rust bad channels: ['F3']

Interpretation

IRASA fitted aperiodic descriptors are now at numerical parity on this AKNL slice.

FOOOF overall parity is high, with auc_FOOOF remaining the largest FOOOF-specific discrepancy.

MIM parity is near exact, but MIC remains sensitive to spatial-filter orientation and remains the main connectivity outlier.

Preprocessing snapshots are included with shared visual scaling and overlays so the remaining GEDAI/filter/interpolation waveform differences are visible.

Included Plot Files

- feature_family_parity.png

- feature_scatter_parity.png

- connectivity_metric_parity.png

- preprocess_snapshot_stacked.png

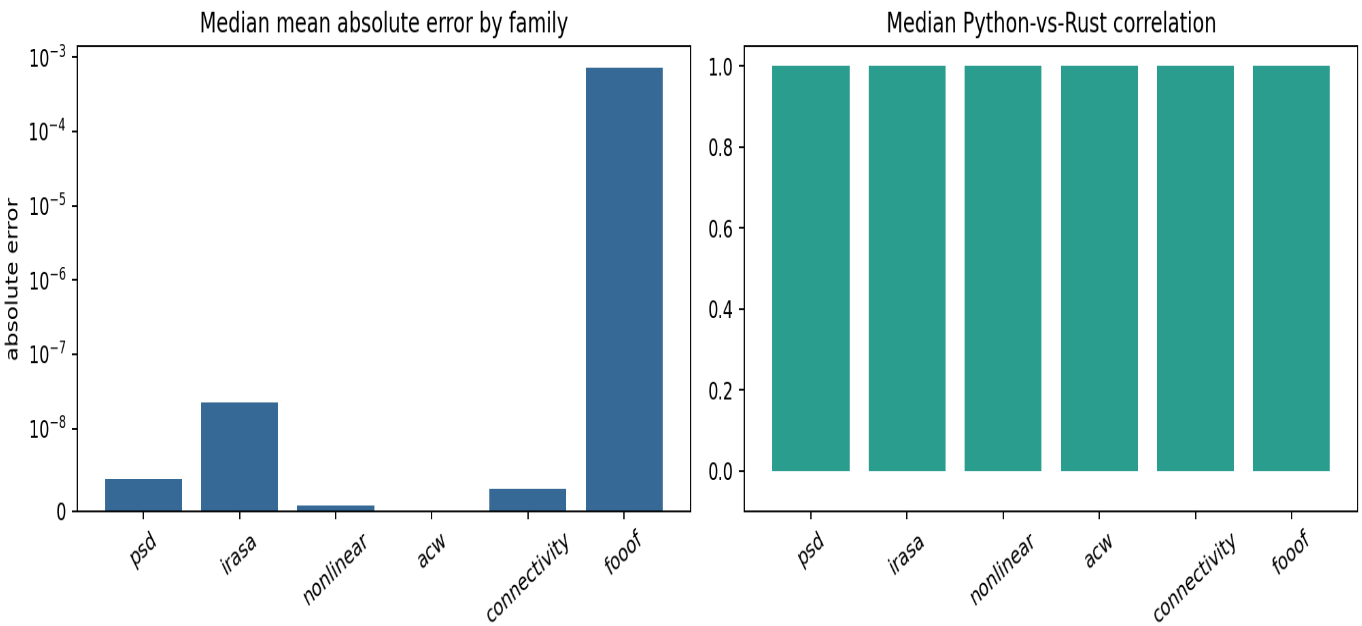
- preprocess_overlay_Fz.png

- preprocess_overlay_F3.png

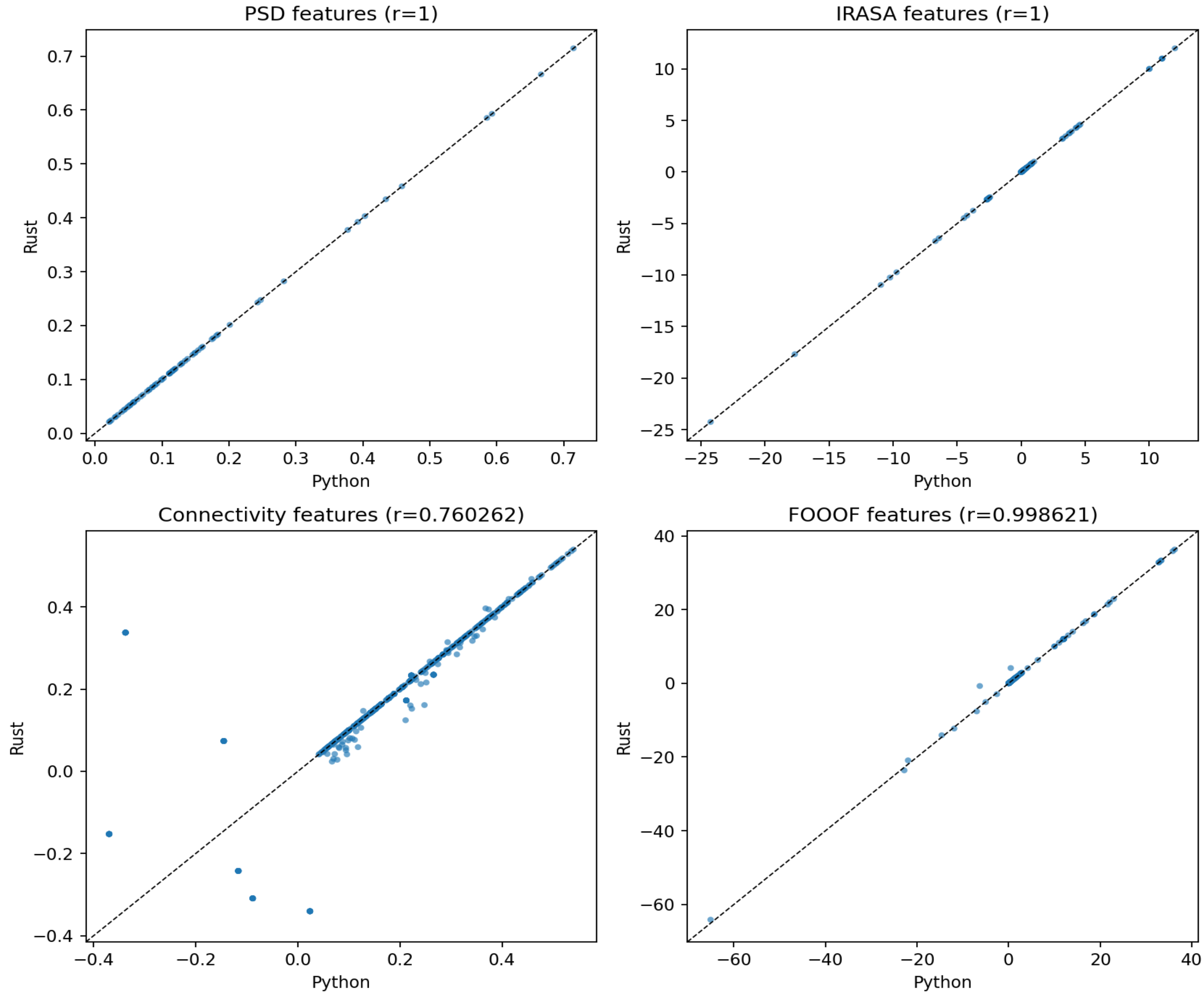
- preprocess_overlay_F4.png

Feature Parity Plots

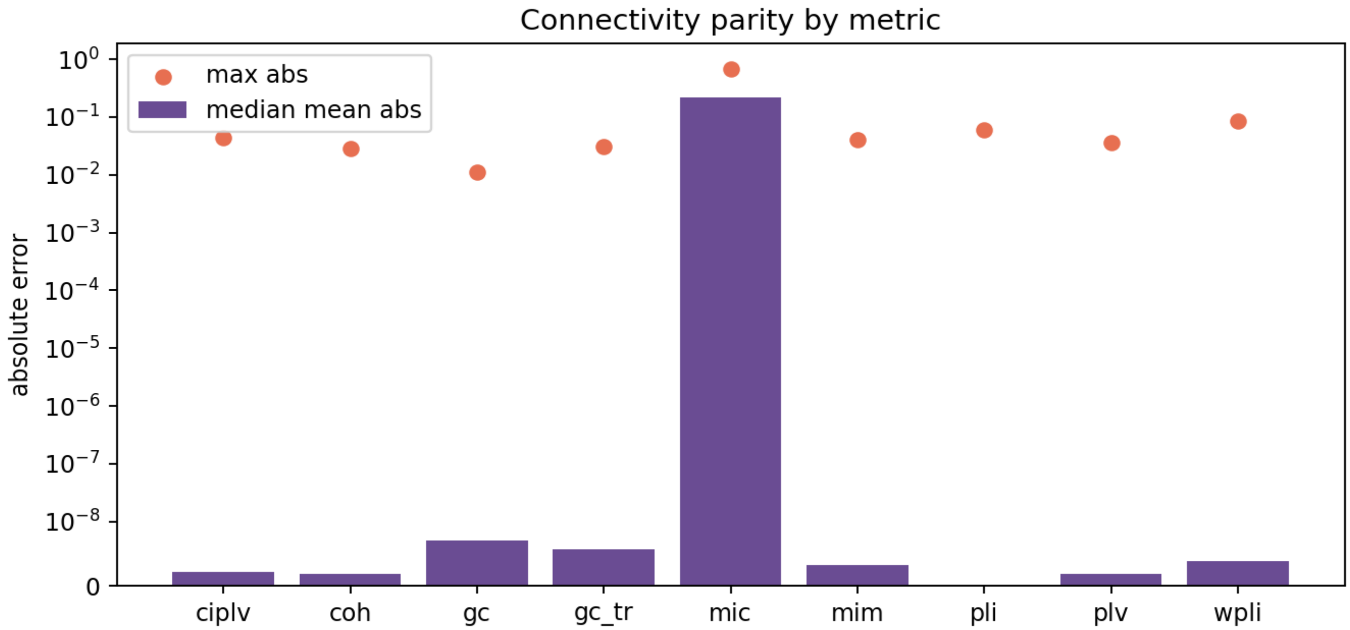
feature_family_parity.png



feature_scatter_parity.png

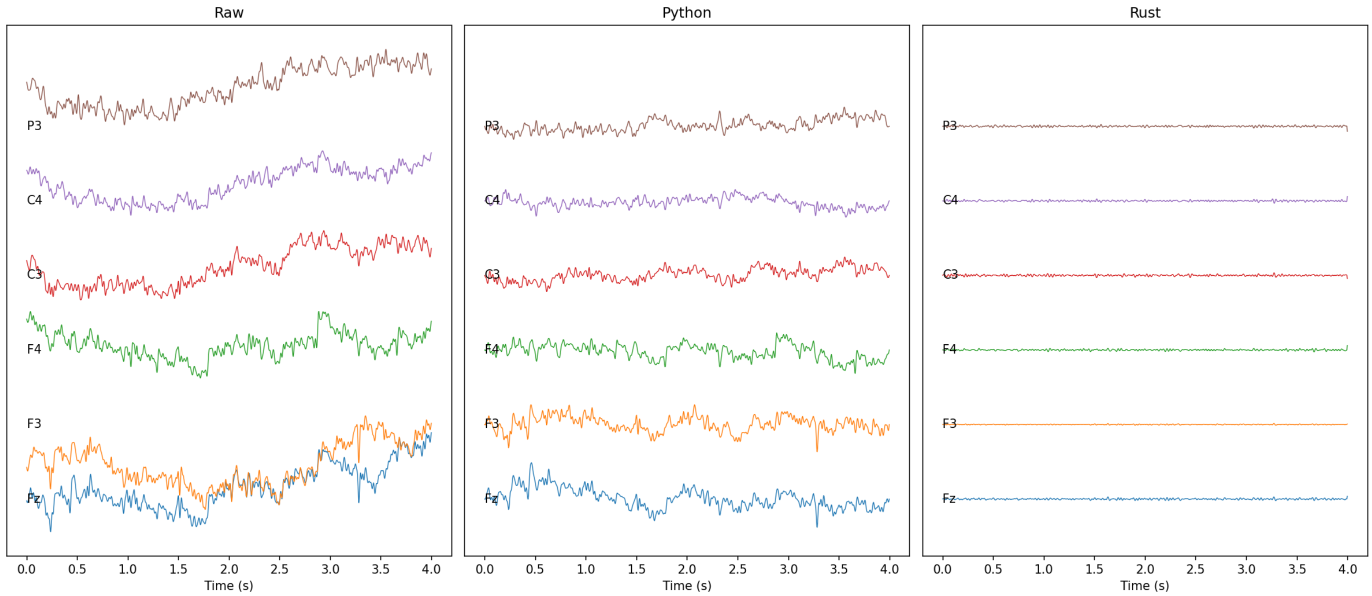


connectivity_metric_parity.png

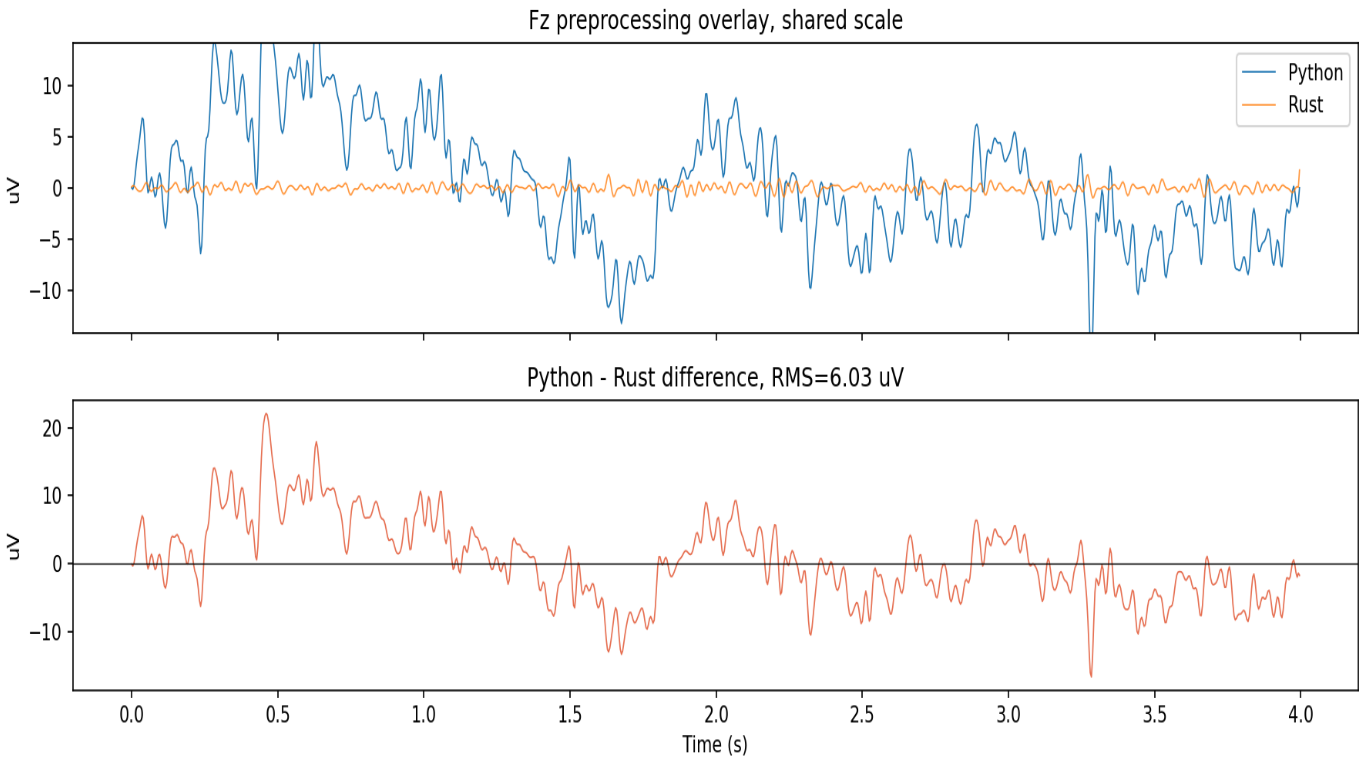


Preprocessing Snapshots

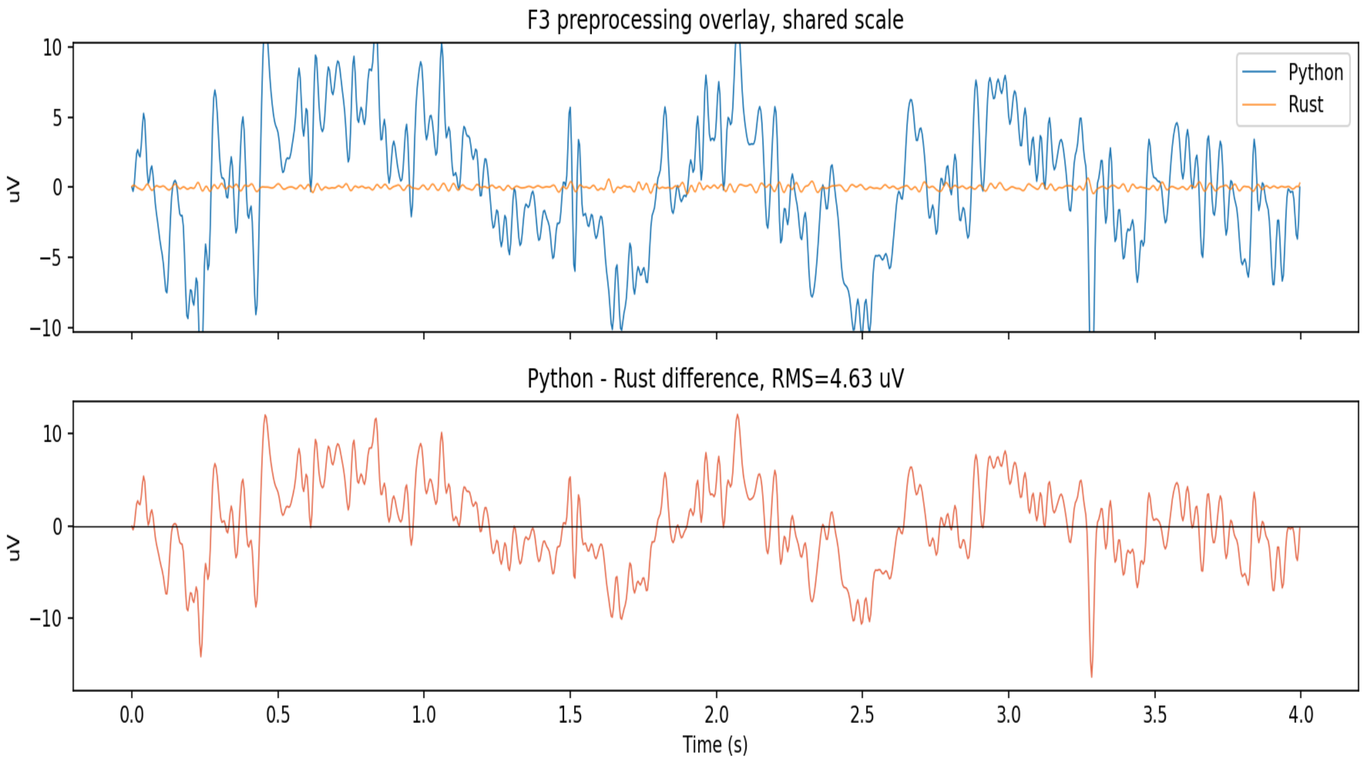
preprocess_snapshot_stacked.png



preprocess_overlay_Fz.png



preprocess_overlay_F3.png



preprocess_overlay_F4.png